BULGARIAN NATIONAL OLYMPIAD IN MATHEMATICS

Fourth round

1996

Problem 1. Find all primary numbers p and q, for which $\dfrac{(5^p-2^p)(5^q-2^q)}{pq}$ is an integer.

Problem 2. Find the side length of the smallest equilateral triangle in which three disks with radii 2, 3 and 4 without common inner points can be placed.

Problem 3. The quadratic functions $f(x)$ and $g(x)$ are with real coefficients and have the following property: if the number $g(x)$ is integer for a positive x, then the number $f(x)$ is integer too. Prove that there are such integers m and n, that $f(x) = mg(x) + n$ for all real x.

Problem 4. The sequence $\{a_n\}_{n=1}^{\infty}$ is defined by

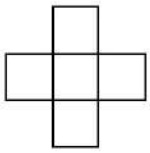
$$a_1 = 1, \quad a_{n+1} = \frac{a_n}{n} + \frac{n}{a_n}, \quad n \geq 1.$$

Prove that $\lfloor a_n^2 \rfloor = n$ when $n \geq 4$ (it is denoted by $\lfloor x \rfloor$ the integer part of the number x).

Problem 5. The quadrilateral $ABCD$ is inscribed in a circle. The lines AB and CD meet each other in the point E, while the diagonals AC and BD — in the point F. The circumcircles of the triangles AFD and BFC have a second common point, which is denoted by H. Prove that $\angle EHF = 90°$.

Problem 6. A square table of size 7×7 with the four corner squares deleted is given.

Figure 4.



a) What is the smallest number of squares which need to be colored black so that a 5-square entirely uncolored Greek cross (Figure 4) cannot be found on the table?

b) Prove that it is possible to writte integers in each square in a way that the sum of the integers in each Greek cross is negative while the sum of all integers in the square table is positive.

Downloaded from Klasirane.com